Julius M. Pfadt

Postdoctoral Researcher (University of Amsterdam) — Co-founder JASP Services BV

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About

I am a researcher in the lab of Eric-Jan Wagenmakers at the University of Amsterdam and a former DFG Walter-Benjamin fellow. My research centers on (Bayesian) statistical modeling, in particular, psychometrics, reliability estimation, and structural equation modeling. I develop open-source tools for JASP (BFpack, Factor, Reliability, SEM) and the R package Bayesrel, and I co-founded JASP Services BV. I am committed to open science and accessible methodology.

Education

2018–2023 **Doctorate (summa cum laude)**, Department of Psychological Research Methods, Ulm University
Project title: *The Present and Future of Reliability Analysis: Advances in Theory and Practice* (Link)
Supervisor: Prof. Dr. Morten Moshagen

2016–2018 **Master of Science**, Psychology, Ulm University, Germany

2012–2016 **Bachelor of Science**, Psychology, Ulm University, Germany

Experience

2025–present **Co-Founder of JASP Services BV**

2024–present **Postdoctoral Researcher**, Programme Group Psychological Methods, University of Amsterdam, The Netherlands

2023 **Postdoctoral Researcher**, Department of Methodology and Statistics, Tilburg University, The Netherlands

2019–present **Developer and Maintainer**, JASP, University of Amsterdam, The Netherlands
Design, implementation, and maintenance of statistical modules

2017 **Research Intern**, Programme Group Psychological Methods, University of Amsterdam, The Netherlands
Topic: Bayesian statistics
Supervisor: Prof. Dr. Eric-Jan Wagenmakers

Publications

2026 **Pfadt, J. M.**, Merkle, E. C., & Wagenmakers, E.-J. (2026). *Bayes factors for structural equation models with blavaan and bridge sampling*. PsyArXiv. https://doi.org/10.31234/osf.io/pt2bc_v2

Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., Mensink, L., Nak, J., De Ruiter, J. P., Sarafoglou, A., Siepe, B. S., Akrong, E., Arena, G., Aust, F., van den Bergh, D., Brenner, W., Doekemeijer, R. A., Donzallaz, M. C., van Doorn, J.,

Echevarria, N. O., . . . Wagenmakers, E.-J. (2026). *The prevalence of Bayesian inference, preregistration, and open data in psychology, 2004-2024*. PsyArXiv. https://doi.org/10.31234/osf.io/ck3js_v3

Mulder, J., & Pfadt, J. (2026). Going in the right direction: A tutorial to directional hypothesis testing using the BFpack module in JASP. *Advances in Methods and Practices in Psychological Science*, *9* (3). https://doi.org/10.1177/25152459261463155

Pfadt, J. M., Merkle, E. C., & Wagenmakers, E.-J. (2026). *Bayes factors for structural equation models with bridge sampling and blavaan*. PsyArXiv. https://doi.org/10.31234/osf.io/pt2bc_v1

Godmann, H. R., Molenaar, D., Ziegler, M., & **Pfadt, J. M**. (2026). *A tutorial on assessing measurement invariance with moderated (non-)linear factor analysis in JASP*. PsyArXiv. https://doi.org/10.31234/osf.io/6ftqg_v2

Pfadt, J. M., Molenaar, D., Hurks, P., & Sijtsma, K. (2026). A tutorial on estimating the precision of individual test scores for anyone constructing and using psychological tests. *Psychometrika*, 1–22. https://doi.org/10.1017/psy.2026.10081

2025 **Pfadt, J. M.**, Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., Mensink, L., Nak, J., De Ruiter, J. P., Sarafoglou, A., Siepe, B. S., Arena, G., Akrong, E., Aust, F., van den Bergh, D., Brenner, W., Doekemeijer, R. A., Donzallaz, M. C., van Doorn, J., Echevarria, N. O., . . . Wagenmakers, E.-J. (2025). *A methodological metamorphosis: The rapid rise of Bayesian inference and open science practices in psychology*. PsyArXiv. https://doi.org/10.31234/osf.io/ck3js_v1

Mulder, J., **Pfadt, J. M.**, & Wagenmakers, E.-J. (2025). A tutorial on Bayesian hypothesis testing of correlation coefficients using the BFpack-module in JASP. *Behavior Research Methods*, *57* (11), 311. https://doi.org/10.3758/s13428-025-02846-5

2023 **Pfadt, J. M.**, van den Bergh, D., & Moshagen, M. (2023). Classical and Bayesian uncertainty intervals for the reliability of multidimensional scales. *Structural Equation Modeling: A Multidisciplinary Journal*, *30* (3), 349–363. https://doi.org/10.1080/10705511.2022.2124162

Pfadt, J. M. (2023), *The present and future of reliability analysis: Advances in theory and practice* (Doctoral dissertation, Ulm University). https://doi.org/10.18725/OPARU-49700

Sijtsma, K., & **Pfadt, J. M**. (2023). Reliability. In R. Tierney, F. Rizvi, & K. Ercikan (Eds.), *International encyclopedia of education* (4th ed., pp. 21-34). Elsevier. https://doi.org/10.1016/B978-0-12-818630-5.10004-1

2022 **Pfadt, J. M.**, van den Bergh, D., Sijtsma, K., Moshagen, M., & Wagenmakers, E.-J. (2022). Bayesian estimation of single-test reliability coefficients. *Multivariate Behavioral Research*, *57* (4), 620–641. https://doi.org/10.1080/00273171.2021.1891855

Pfadt, J. M., van den Bergh, D., Sijtsma, K., & Wagenmakers, E.-J. (2022). A tutorial on Bayesian single-test reliability analysis with JASP. *Behavior Research Methods*, *55* (3), 1069–1078. https://doi.org/10.3758/s13428-021-01778-0

Pfadt, J. M., & Sijtsma, K. (2022). Statistical properties of lower bounds and factor analysis methods for reliability estimation. In *Quantitative Psychology* (pp. 51–63). Springer International Publishing. https://doi.org/10.1007/978-3-031-04572-1_5

2021 Sijtsma, K., & **Pfadt, J. M**. (2021). Rejoinder: The future of reliability. *Psychometrika*, *86* (4), 887–892. https://doi.org/10.1007/s11336-021-09807-9

Sijtsma, K., & **Pfadt, J. M**. (2021). Part II: On the use, the misuse, and the very limited usefulness of Cronbach's alpha: Discussing lower bounds and correlated errors. *Psychometrika*, *86* (4), 843–860. https://doi.org/10.1007/s11336-021-09789-8

Software

2019–present **Bayesrel**, R-package
Creator and maintainer of the R-package for Bayesian reliability estimation

2020–present **Modules in JASP: BFpack, Factor, Reliability, SEM**, Statistics program
Maintainer

Teaching

2024, 2026 **Structural Equation Modelling**, Lecturer, Research master course, University of Amsterdam, The Netherlands

2016–2019, 2021 **Applications of Multivariate Statistics in R**, Instructor, Master course, Ulm University, Germany

Talks

Conference **European Network for Business and Industrial Statistics (ENBIS) Conference 2026**, "JASP for Quality Control: An Open-Source Software for Industrial Statistics", Florence, Italy

Conference **International Meeting of the Psychometric Society (IMPS) 2026**, "Bayes Factors for Structural Equation Models with blavaan and bridgesampling", Seoul, South Korea

Conference **Association for Psychological Science (APS) 2026 Annual Convention**, "The Past, Present, and Future of Scientific Reform", Barcelona, Spain

Conference **International Meeting of the Psychometric Society 2021**, "Bayesian Multidimensional Reliability", online

Invited **Research Section of the Forensic Psychiatry**, "Bayesian statistics", Guenzburg, Germany

Funding, Awards & Scholarships

2024–2026 **Walter-Benjamin Fellowship** from the German Research Foundation (DFG) for "Bayesian Model Averaging for Structural Equation Models"

2025 **Open-Science Award** from the Open Science Community Amsterdam for "Tracking Change in Psychology: A Meta Research Project Using Human Coders and AI"

Reviewing

Ad-hoc reviewer for: *Behavior Research Methods*, *BMC Medical Research Methodology*, *BMC Musculoskeletal Disorders*, *British Journal of Mathematical and Statistical Psychology*, *Educational and Psychological Measurement*, *European Journal of Psychological Assessment*, *German Journal of Exercise and Sport Research*, *Journal of Mathematical Psychology*, *Perspectives on Psychological Science*, *Psychological Methods*, *Psychometrika*, *Structural Equation Modeling: A Multidisciplinary Journal*